# BT 3041 Analysis and Interpretation of Biological Data

## Assignment 2

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All the relevant code for the problems solved as part of this assignment can be found here:
https://colab.research.google.com/drive/1IBQiprDX1Eqdu-l4tJsUFtoEsAE15WkN?usp=sharing

The Google Drive folder containing all relevant data, code and the report can be found here:

https://drive.google.com/drive/folders/18PUegvMmaqVf2eQnzqJyAu1uFnlMR40-?usp=sharing

1. **Section 1- Regression:**

   a. In order to split the dataset into Training, Validation and Testing data, we first split the dataset such that 20% of the data is labelled as Testing data, and 25% of the rest (which is 20% of the total) is labelled as the Validation data; the remaining 80% of the total data is labelled as Training data. The size of each of these datasets is given as follows:

| Type of Data | Size of the Dataset |
|---|---|
| 1. Training Data | 994 data points |
| 2. Validation Data | 332 data points |
| 3. Testing Data | 332 data points |

Note: The values of the feature variables were normalised with respect to each other to account for the different ranges of values of each of them, and make meaningful conclusions using the coefficients of linear regression that we aimed to obtain. Since the feature variables were normalised, it was deemed necessary to normalise the values of the predictor variable as well.

b.

- The plot of RSS values against the features included in the model is given as follows:

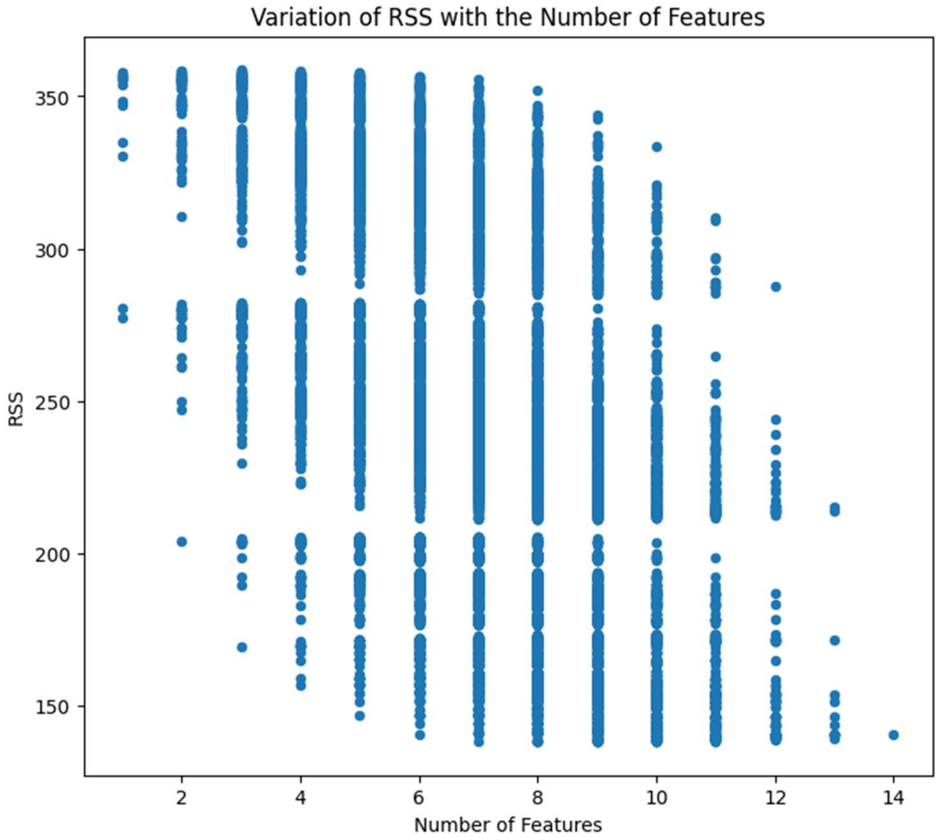


Figure 1: Variation of RSS values with the Number of Features

- The 3 models with the lowest AIC values are given as follows:

| Number of features | Features | RSS | AIC | Intercept | Coefficients |
|---|---|---|---|---|---|
| 7 | ['N', 'P', 'K', 'Soil_Moisture', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used'] | 138.4086537 | 1850.637268 | 0.000879692 | [0.13124817551989007, 0.0871919239870275, 0.12668571174663415, 0.26559024197570646, 0.42680443523851597, -0.14904264971771547, 0.5164641540230721] |
| 8 | ['N', 'P', 'K', 'Soil_Moisture', 'Temperature', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used'] | 138.5471703 | 1850.874154 | 0.000815138 | [0.13158871559694307, 0.08785187409663243, 0.12713881860211357, 0.26541076150133686, -0.025738937373604692, 0.4269730116871225, -0.14970022789365542, 0.5149787188373551] |
| 8 | ['N', 'P', 'K', 'Soil_Moisture', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used', 'Pesticide_Used'] | 139.7286996 | 1851.349983 | 0.001163664 | [0.1304800414074655, 0.08629523821653864, 0.1260945597134275, 0.2659367361970022, 0.42613792507509846, -0.14995534615655917, 0.5166171697684181, -0.022466304421435636] |

Figure 2: Top 3 Linear Regression models with the lowest AIC values

c. Upon performing 5-fold cross-validation for the top 3 models obtained below, the mean and variance of the MSE's during the cross-validation were obtained as follows:

| Number of features | Features | RSS | AIC | Mean of MSE | Variance of MSE |
|---|---|---|---|---|---|
| 7 | ['N', 'P', 'K', 'Soil_Moisture', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used'] | 138.4086537 | 1850.637268 | 0.378889043 | 0.001188477 |
| 8 | ['N', 'P', 'K', 'Soil_Moisture', 'Temperature', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used'] | 138.5471703 | 1850.874154 | 0.379544618 | 0.001200102 |
| 8 | ['N', 'P', 'K', 'Soil_Moisture', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used', 'Pesticide_Used'] | 139.7286996 | 1851.349983 | 0.378929672 | 0.001275271 |

Figure 3: Mean and Variance of MSE values obtained after performing 5-fold cross validation

- The top 3 models do not show a wide variation in their values of RSS, AIC and mean MSE. The relatively low magnitude of MSE values for each of these models indicates that they are stable, and did not generate widely different values of MSE for different folds of the training dataset that were used for training and testing during cross-validation.

d. The performance of the top 3 models, as given above, on the validation dataset is given as follows:

| Number of features | Features | RSS | AIC | RMSE | MAE | $R^2$ |
|---|---|---|---|---|---|---|
| 7 | ['N', 'P', 'K', 'Soil_Moisture', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used'] | 138.4086537 | 1850.637268 | 0.624327959 | 0.491111253 | 0.60776643 |
| 8 | ['N', 'P', 'K', 'Soil_Moisture', 'Temperature', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used'] | 138.5471703 | 1850.874154 | 0.626043166 | 0.493069165 | 0.605608315 |
| 8 | ['N', 'P', 'K', 'Soil_Moisture', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used', 'Pesticide_Used'] | 139.7286996 | 1851.349983 | 0.626220947 | 0.493371552 | 0.605384288 |

Figure 4: Performance of the top 3 models on the validation dataset

- All the top 3 models report the same value of RMSE, MAE and $R^2$, indicating that the additional $8^{th}$ parameter in two of the models (Temperature or Pesticide Used) do not substantially contribute to improving the values of the error metrics.
- Thus, based on these error metrics, N, P, K, Soil Moisture, Rainfall, Wind Speed and Fertiliser Used can be considered as important feature variables that help relatively accurately predict the target variable, at least in the validation stage of constructing the model.
- The value of $R^2$ is approximately equal to 0.6 in each of the three cases, which is lower than the desirable value (above 0.8 or 0.9). Therefore, other forms of regression such as non-linear (for instance, polynomial) regression can be explored to determine if some of the predictor variables affect the target variable in a non-linear manner.

e. L1- and L2-regularisation models were constructed by setting the penalty parameter (otherwise denoted as λ) α arbitrarily to demonstrate how the choice of α is instrumental in determining the magnitude of error metrics like RMSE, MAE and $R^2$.

These models were trained on the Training data and used to predict the Validation data and generate error metrics. These error metrics shall then be used to compare with the ones obtained in part (g) after using an optimal value of α as obtained from 5-fold cross validation. Cross validation was performed using the Training data in each case.

The performance of each of these models on the validation dataset is given as follows:

```
The following model parameters were obtained upon setting the parameter lambda = 0.5 :

Intercept =  -0.008234073633902142
Betas =  [ 0.          0.          0.         -0.          0.         -0.
 -0.          0.          0.          0.         -0.          0.
  0.03706167 -0.        ]

The following error metrics were obtained upon setting the parameter lambda = 0.5 :
RMSE =  0.9768199964699404
MAE =  0.806762533432434
R2 =  0.03982968555444655

The cross validation score obtained upon setting the parameter lambda = 0.5 :
Cross Validation Score =  0.02001347596884908 with standard deviation =  0.03196663379204248
```

Figure 5: Model Parameters and Error Metrics for the L1-regularised model on predicting the target variable using the Validation dataset

```
The following model parameters were obtained upon setting the parameter lambda = 10000 :

Intercept =  -0.007086695614777875
Betas =  [ 0.01321964  0.00848489  0.00914136 -0.00323208  0.0222705  -0.0024152
 -0.00389019  0.0011029   0.04184554  0.00188814 -0.01436397  0.00218612
  0.04828622 -0.00289949]

The following error metrics were obtained upon setting the parameter lambda = 10000
RMSE =  0.9390530113113621
MAE =  0.7776256219950404
R2 =  0.11264089218363216

The cross validation score obtained upon setting the parameter lambda = 10000 :
Cross Validation Score =  0.07466213448862098 with standard deviation =  0.017117354401736992
```

Figure 6: Model Parameters and Error Metrics for the L2-regularised model on predicting the target variable using the Validation dataset

f.

- Regularisation was performed using the LassoCV() and RidgeCV() functions provided by the scikit-learn package, with default parameters.
- There are other ways in which the user can specify which values of α to consider, and thereby search the α parameter search, such as using the GridSearchCV() function in the scikit-learn package.
- Here, the value of $R^2$ was used as the scoring metric within the LassoCV() and RidgeCV() functions.
- In the case of L1-regularisation, the optimal λ was chosen as:

| | |
|---|---|
| λ | 0.0057524173079022654 |

- In case of L2-regularisation, the optimum λ was chosen as:

| | |
|---|---|
| λ | 10 |

g.

- The performance of the L1-regularisation model with the optimal value of λ on the validation dataset is given as follows:

```
The following model parameters were obtained upon setting the parameter lambda = 0.0057524173079022654 :

Intercept =  0.001683843122741828
Betas =  [ 0.12706829  0.08016769  0.12003121 -0.00474093  0.25913569 -0.0111868
 -0.01980948 -0.01017672  0.42187654  0.01185878 -0.14357087  0.01481357
  0.50951464 -0.01734041]

The following error metrics were obtained upon setting the parameter lambda = 0.0057524173079022654 :
RMSE =  0.627833978919744
MAE =  0.49334207921657325
R2 =  0.6033487523939571

The cross validation score obtained upon setting the parameter lambda = 0.0057524173079022654 :
Cross Validation Score =  0.5984434636122501 with standard deviation =  0.05087571934981118
```

Figure 7: Model Parameters and Error Metrics for the L1-regularisation model with λ optimal on predicting the target variable using the Validation data

- We note that the values of the error metrics such as RMSE and MAE have markedly reduced compared to the ones obtained using the arbitrary λ, while the value of $R^2$ has substantially improved, and is close the one obtained by using the any one of the top 3 models based on AIC as given above.
- The relatively high magnitudes of the coefficients used for N, K, Soil Moisture, Rainfall and Fertiliser Used indicate that these features substantially impact the yield, and bear positive correlation with the target variable. All the features that correlate negatively with the target variable have relatively lower coefficients, indicating that they do not substantially impact the crop yield.

- The performance of the L2-regularisation model with the optimal value of λ on the validation dataset is given as follows:

```
The following model parameters were obtained upon setting the parameter lambda = 10.0 :

Intercept =  0.002477047607146182
Betas =  [ 0.13245736  0.08458604  0.12496017 -0.01019398  0.26205207 -0.01684006
 -0.02572756 -0.01609451  0.42304071  0.01700133 -0.14808981  0.02025703
  0.50963942 -0.02322815]

The following error metrics were obtained upon setting the parameter lambda = 10.0
RMSE =  0.6305336155624514
MAE =  0.49475625206298
R2 =  0.5999302803555497

The cross validation score obtained upon setting the parameter lambda = 10.0 :
Cross Validation Score =  0.5973251912342707 with standard deviation =  0.05080174078749018
```

Figure 8: Model Parameters and Error Metrics for the L2-regularisation model with λ optimal on predicting the target variable using the Validation data

- We note that the values of the error metrics such as RMSE and MAE have markedly reduced compared to the ones obtained using the arbitrary λ, while the value of $R^2$ has substantially improved, and is close the one obtained by using the any one of the top 3 models based on AIC as given above. This agrees with the observations we obtained from performing L1-regularisation.
- The relatively high magnitudes of the coefficients used for N, K, Soil Moisture, Rainfall and Fertiliser Used indicate that these features substantially impact the yield, and bear positive correlation with the target variable. All the features that correlate negatively with the target variable have relatively lower coefficients, indicating that they do not substantially impact the crop yield. These observations match with the ones we made after performing L1-regularisation.

- However, the values of the error metrics obtained from L1- and L2-regularisation are close to each other, indicating there is no substantial difference in the performance of the model on the validation dataset regardless of the type of regularisation used.

h. The performance of the different models built above on the test data is given as follows:

| Number of features | Features | RSS | AIC | RMSE | MAE | R2 |
|---|---|---|---|---|---|---|
| 7 | ['N', 'P', 'K', 'Soil_Moisture', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used'] | 138.4086537 | 1850.637268 | 0.645672932 | 0.516770525 | 0.609637404 |
| 8 | ['N', 'P', 'K', 'Soil_Moisture', 'Temperature', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used'] | 138.5471703 | 1850.874154 | 0.645995939 | 0.51728893 | 0.609246737 |
| 8 | ['N', 'P', 'K', 'Soil_Moisture', 'Rainfall', 'Wind_Speed', 'Fertilizer_Used', 'Pesticide_Used'] | 139.7286996 | 1851.349983 | 0.648744616 | 0.517866116 | 0.605914396 |

Figure 9: Performance of the top 3 (based on AIC) models on the Training dataset

```
The following error metrics were obtained upon setting the parameter lambda = 0.0057524173079022654 :
RMSE =  0.6509070178177412
MAE =  0.5177850087913498
R2 =  0.6032828777405627
```

Figure 10: Performance of the L1-regularisation model on the Training dataset

```
The following error metrics were obtained upon setting the parameter lambda = 10 :
RMSE =  0.651498463895065
MAE =  0.5170143250328061
R2 =  0.6025615968798741
```

Figure 11: Performance of the L2-regularisation model on the Training dataset

- Based on the above figures, the models developed in parts (d) and (g) perform similarly on the test data, since the values of all relevant error metrics are observed to be close to each other, regardless of the model or form of regularisation used, indicating stability and less variance across different models.
- However, we note that the issue pertaining to the relatively high the value of $R^2$ was not eliminated upon performing regularisation.

i. Upon considering the models obtained in parts (d) and (g) we note that they perform similarly across validation and test data, since the values of all relevant error metrics are observed to be close to each other. The performance of the models can be further improved by considering the following:

1. Splitting the data in a different manner, say by using a different random seed, and observe if these results are reproduced.
2. Searching the parameter space for the hyper-parameter $\lambda$ more extensively to obtain regularisation models that reduce the coefficients of different feature variables

to zero and improve accuracy, i.e. reduce the values of error metrics when Test data is predicted.

3. Exploring non-linear, such as polynomial, regression models to best capture the effect of certain feature variables on the target variable, even though this may be computationally intensive, and, therefore, must be informed by literature review.

N, P, K, Soil Moisture, Rainfall, Wind Speed and Fertiliser Used were the top 7 feature variables identified in the model with the lowest AIC, as tabulated above, indicating they are important in predicting the target variable accurately. Furthermore, these variables (barring P) were observed to have positive coefficients with a relatively high magnitude upon performing regularisation, thus underlining their importance.

However, these variables may positively correlate with crop yield only in a certain range. For example, heavy rainfall and extensive fertiliser use can be detrimental to crops. Therefore, data pertaining to excessive use of N, P, K minerals, fertiliser dose, etc. must be considered to optimally manager these important variables in order to ensure optimal crop yield.

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2. **Section 2 - Classification:**

a. In order to split the dataset into Training, Validation and Testing data, we first split the dataset such that 20% of the data is labelled as Testing data, and 25% of the rest (which is 20% of the total) is labelled as the Validation data; the remaining 80% of the total data is labelled as Training data. The size of each of these datasets is given as follows:

| Type of Data | Size of the Dataset |
|---|---|
| 4. Training Data | 994 data points |
| 5. Validation Data | 332 data points |
| 6. Testing Data | 332 data points |

Note: The values of the feature variables were not normalised, since doing so would require normalising those of the target variable as well; however, since the target variable is a categorical variable here, normalisation was not performed.

b. An arbitrary decision tree classifier was developed using default hyper-parameters, and cross-validation scores were obtained as follows:

| Metric | Mean | Standard Deviation |
|---|---|---|
| Accuracy | 0.541211106 | 0.05457422 |
| Balanced Accuracy | 0.499690047 | 0.053344011 |
| Average Precision | 0.424681549 | 0.034260421 |
| F1_Micro | 0.541211106 | 0.05457422 |
| F1_Macro | 0.495828339 | 0.053133224 |

Figure 12: Cross-Validation scores for an arbitrary decision tree with depth = 14

The decision tree has been visualised in the .ipynb file containing the relevant code.

c.

- Cross-validation was performing using the training set, and considering accuracy and balanced accuracy as two different metrics. Balanced Accuracy was considered since this scoring metric is recommended as the suitable one for imbalanced datasets.
- Cross-validation was performed using the GridSearchCV() function in the scikit-learn package.
- Here, almost 2/3 of the datapoints in the target variable data correspond to a ‘High’ value; thus, the given dataset was considered imbalanced, which warranted the use of the imbalanced accuracy metric.
- The optimal value of the depth of the tree that resulted in the highest accuracy in the given range was found to be 2, with a cross-validation score of 0.5924724633267346.

- The optimal value of the depth of the tree that resulted in the highest balanced accuracy in the given range was found to be 2, with a cross-validation score of 0.5323256651500781.
- The variation of accuracy with depth of the decision tree is plotted as follows:

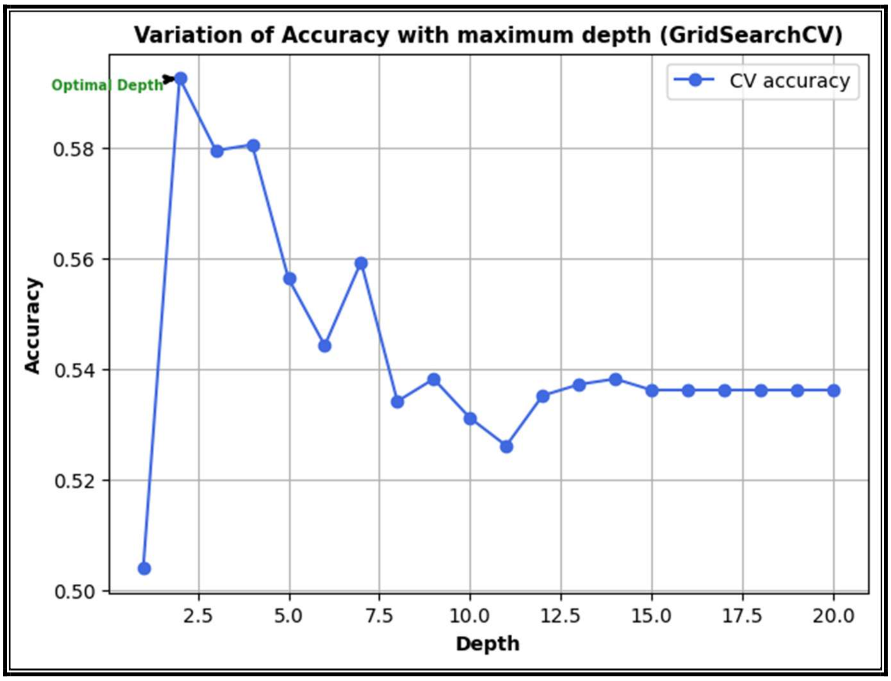


Figure 13: Variation of Accuracy with the Depth of the Decision Tree

- The variation of balanced accuracy with depth of the decision tree is plotted as follows:

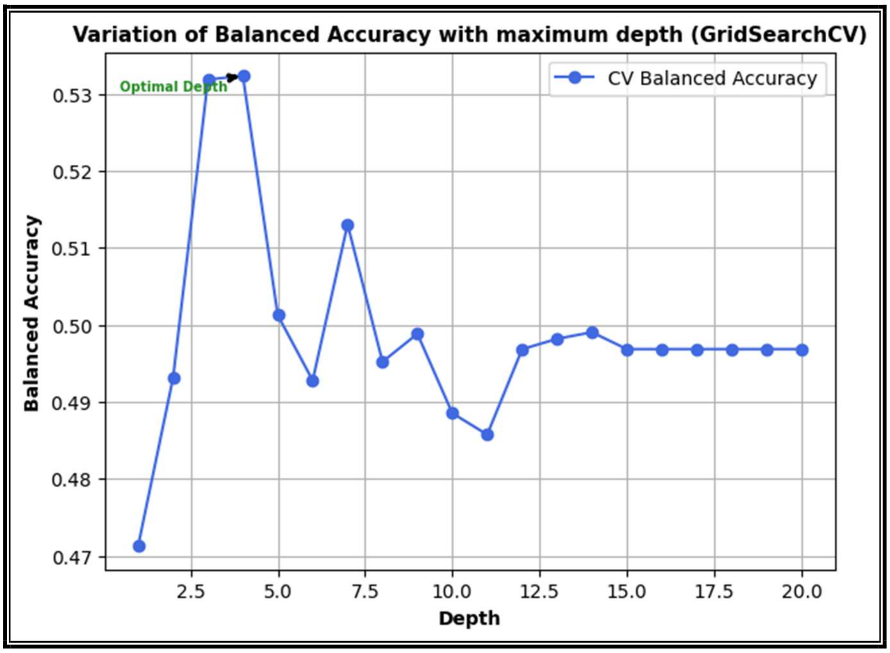


Figure 14: Variation of Balanced Accuracy with the Depth of the Decision Tree

- For the depth of the decision tree = 2, the following cross-validation scores were obtained:

| Metric | Mean | Standard Deviation |
|---|---|---|
| Accuracy | 0.592472 | 0.042155309 |
| Balanced Accuracy | 0.493099 | 0.04528569 |
| Average Precision | 0.5016 | 0.039609855 |
| F1_Micro | 0.592472 | 0.042155309 |
| F1_Macro | 0.424606 | 0.041437866 |

Figure 15: Cross-Validation scores for the decision tree with depth = 2

- For the depth of the decision tree = 2, the following cross-validation scores were obtained:

| Metric | Mean | Standard Deviation |
|---|---|---|
| Accuracy | 0.575392 | 0.040615734 |
| Balanced Accuracy | 0.521939 | 0.039314933 |
| Average Precision | 0.537137 | 0.038952379 |
| F1_Micro | 0.575392 | 0.040615734 |
| F1_Macro | 0.523622 | 0.040544895 |

Figure 15: Cross-Validation scores for the decision tree with depth = 4

- Since both the decision trees perform in a similar way, based on the scoring metrics considered for the cross-validation carried out using Training data, we consider both decision trees for further analyses.
- The decision trees have been visualised in the .ipynb file containing the relevant code.

d.

- The performance of the decision tree with depth = 2 on the Validation dataset is given as follows:

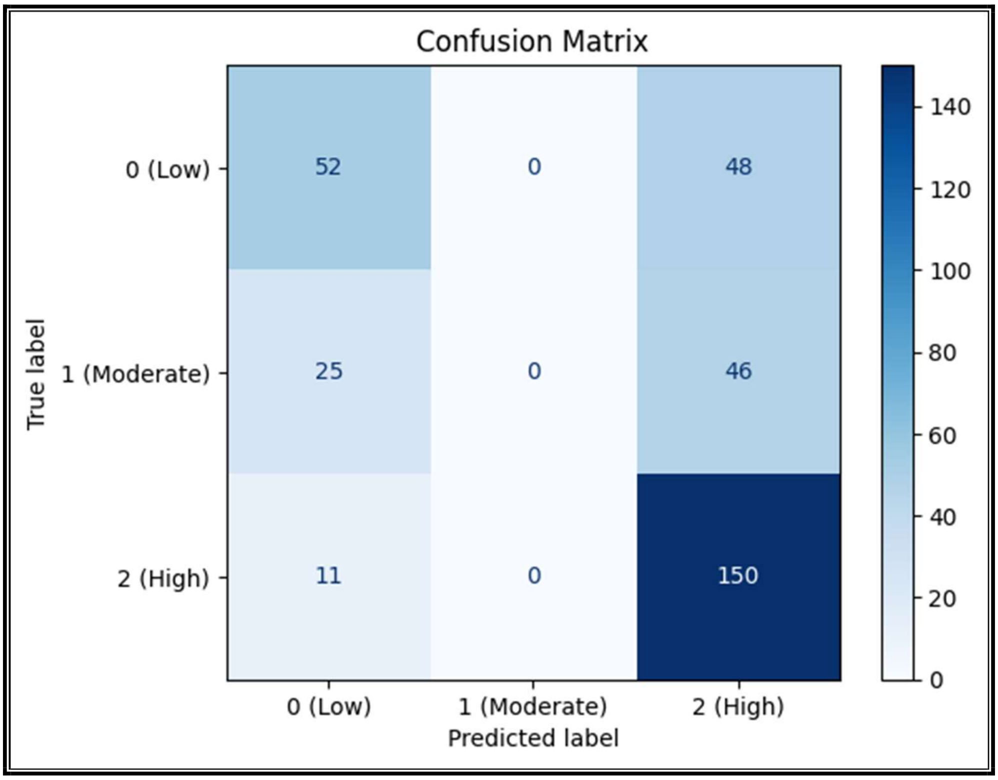


Figure 16: Confusion Matrix for the decision tree with depth = 2

Report :

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.59 | 0.52 | 0.55 | 100 |
| 1 | 0.00 | 0.00 | 0.00 | 71 |
| 2 | 0.61 | 0.93 | 0.74 | 161 |
| accuracy | | | 0.61 | 332 |
| macro avg | 0.40 | 0.48 | 0.43 | 332 |
| weighted avg | 0.48 | 0.61 | 0.53 | 332 |

Figure 17: Classification Report for the decision tree with depth = 2

```
The following metrics are obtained when maximum depth = 2:
Accuracy: 60.8433734939759
Balanced Accuracy: 48.38923395445134
Precision: 47.61033702619719
Recall: 60.8433734939759
F1-Micro: 60.8433734939759
F1-Macro: 43.13107433674809
```

Figure 18: Various Metrics (%) for the decision tree with depth = 2

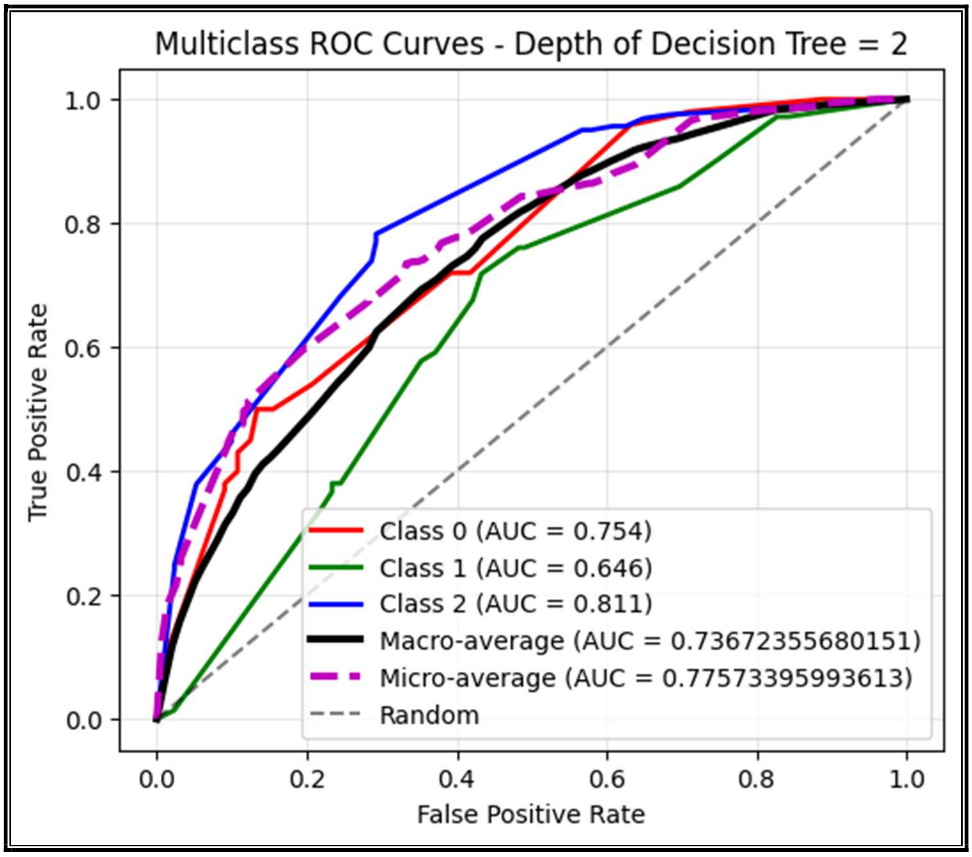


Figure 19: Multiclass ROC curve for the decision tree with depth = 2

- The performance of the decision tree with depth = 4 on the Validation dataset is given as follows:

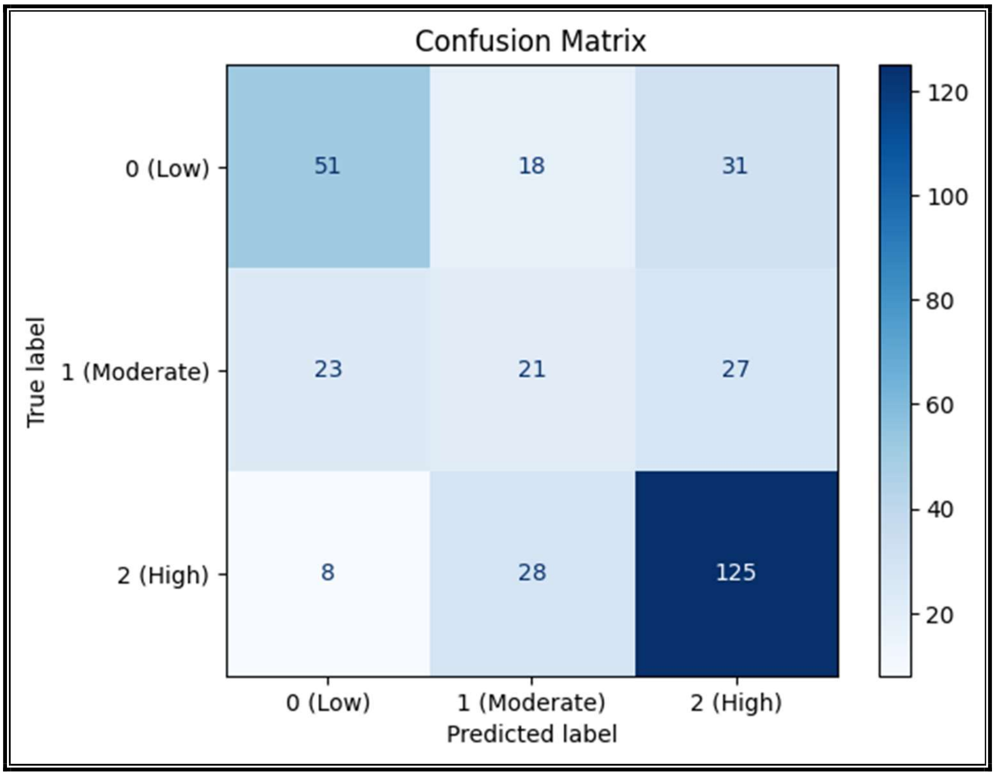


Figure 20: Confusion Matrix for the decision tree with depth = 2

```
Report :
              precision    recall  f1-score   support

           0       0.62      0.51      0.56       100
           1       0.31      0.30      0.30        71
           2       0.68      0.78      0.73       161

    accuracy                           0.59       332
   macro avg       0.54      0.53      0.53       332
weighted avg       0.59      0.59      0.59       332
```

Figure 21: Classification Report for the decision tree with depth = 4

```
The following metrics are obtained when maximum depth = 4:
Accuracy: 59.337349397590366
Balanced Accuracy: 52.739072113842475
Precision: 58.56070207534082
Recall: 59.337349397590366
F1-Micro: 59.337349397590366
F1-Macro: 53.051052419100955
```

Figure 22: Various Metrics (%) for the decision tree with depth = 4

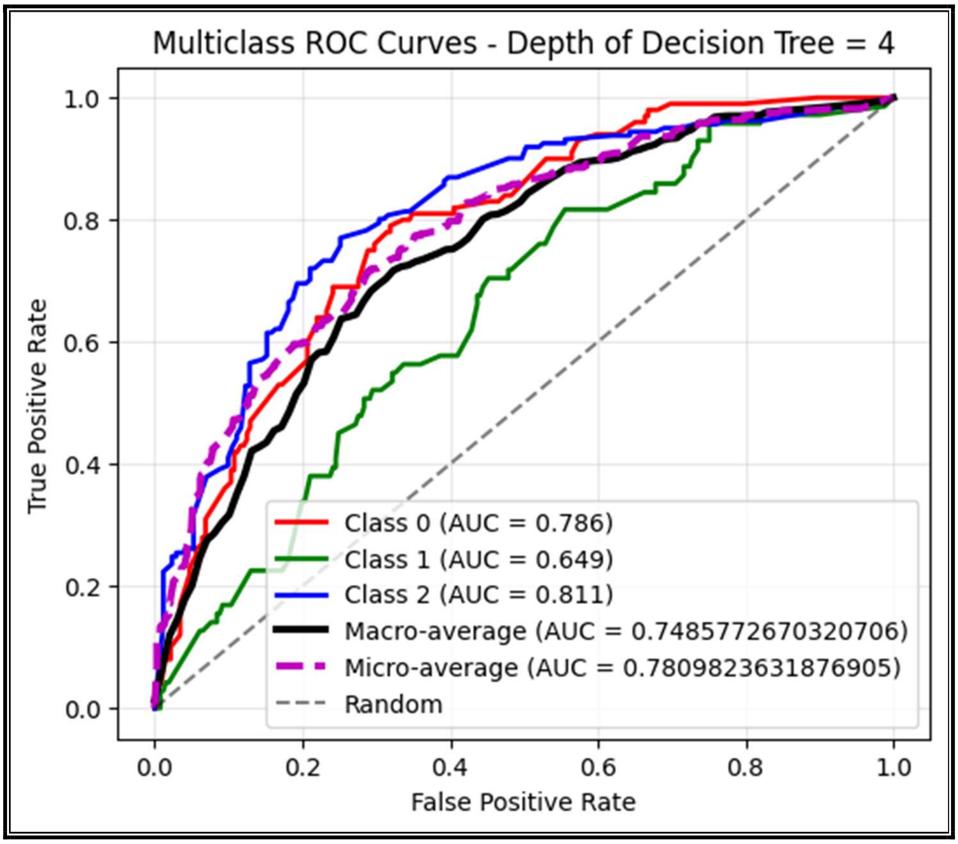


Figure 22: Multiclass ROC curve for the decision tree with depth = 4

- We note that while the tree with depth = 2 reports a higher accuracy, it cannot predict elements to belong to the 'Moderate' category of the target variable. The tree with depth = 4 has a slightly lower accuracy, but predicts elements in all three categories of the target variable.
- The relative values of AUC for Class 0 and Class 1 as compared to that for Class 2 indicate that the decision trees are better at correctly classifying elements that belong to Class 2 ('High') as compared to those in Classes 0 and 1 ('Low' and Moderate'). This could potentially due to the substantial representation of elements belonging to Class 2 ('High') in the given dataset. This is also seen with respect to the low value of Recall for Class 2 in both cases; in fact, Recall = 0 for the decision tree with depth = 2 since the model fails to predict True Positives for elements in Class 2.
- While the Recall value for Class 1 does not change much across these two cases, the Recall value for Class 2 is substantially lower when the depth = 4 as compared to when the depth = 2, which is indicative of both a decrease in the False Positives and a decrease in False Positives (with respect to Class 2).
- Since both the number of True Positives and False Positives for Class 1 approximately remain the same across the two cases, we can conclude that change in those for Class 2 is reflected in the increased number of True Positives and False Positives for Class 3.
- Both the models report average AUC values that are close to 0.75; thus, the decision tree models can be considered to be fairly accurate, since a substantially higher value of AUC (say, close to 0.90) could indicate overfitting.

e.

- An arbitrary kNN classifier was built with k = 20, and the cross-validation scores (cross validation performed using the Training data) for it were reported as follows:

| Metric | Mean | Standard Deviation |
|---|---|---|
| Accuracy | 0.556312 | 0.016739867 |
| Balanced Accuracy | 0.488062 | 0.02983249 |
| Average Precision | 0.502509 | 0.013931871 |
| F1_Micro | 0.556312 | 0.016739867 |
| F1_Macro | 0.46194 | 0.032089144 |

Figure 23: Cross-Validation scores for a kNN classifier with k = 20

- Optimal value of k was determined by plotting the cross-validation score with accuracy as the scoring metric for values of k sampled over an arbitrarily wide range. The value of k that resulted in the highest accuracy score was chosen as the optimal one. The Accuracy Score versus k value plot is given as follows:

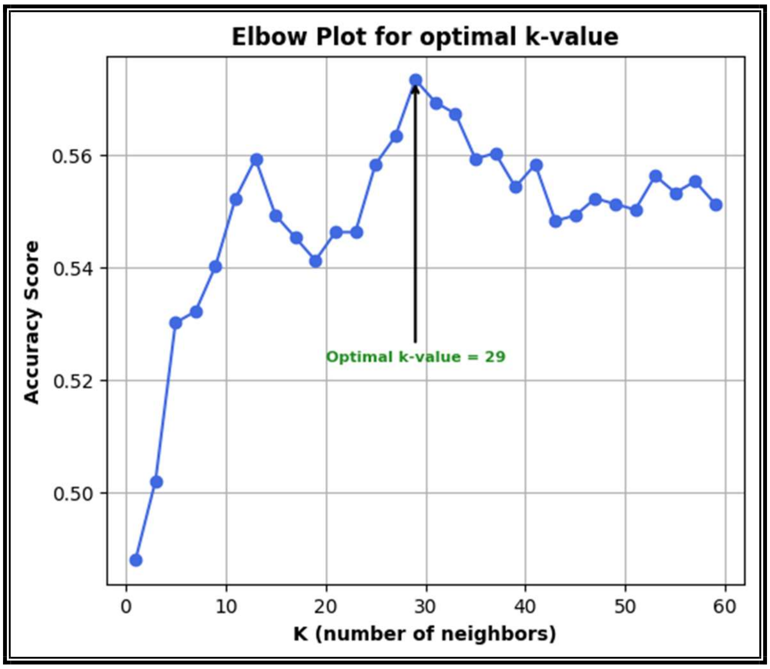


Figure 24: Variation of the Accuracy Score with the k value

- Thus, k = 29 was chosen as the optimal k value for the further analyses.

f. The cross-validation scores for the kNN classifier with k = 29 are given as follows:

| Metric | Mean | Standard Deviation |
|---|---|---|
| Accuracy | 0.573402 | 0.023318275 |
| Balanced Accuracy | 0.506768 | 0.028273249 |
| Average Precision | 0.514152 | 0.015582999 |
| F1_Micro | 0.573402 | 0.023318275 |
| F1_Macro | 0.47595 | 0.035721609 |

Figure 25: Cross-Validation scores for a kNN classifier with k = 29

- We note how the choice of optimal k value improves all the cross-validation scores as compared to an arbitrary choice of k. This difference can be observed to be starker by considering arbitrary k values widely different from the optimal one.

g. The performance of the decision tree with depth = 4 on the Validation dataset is given as follows:

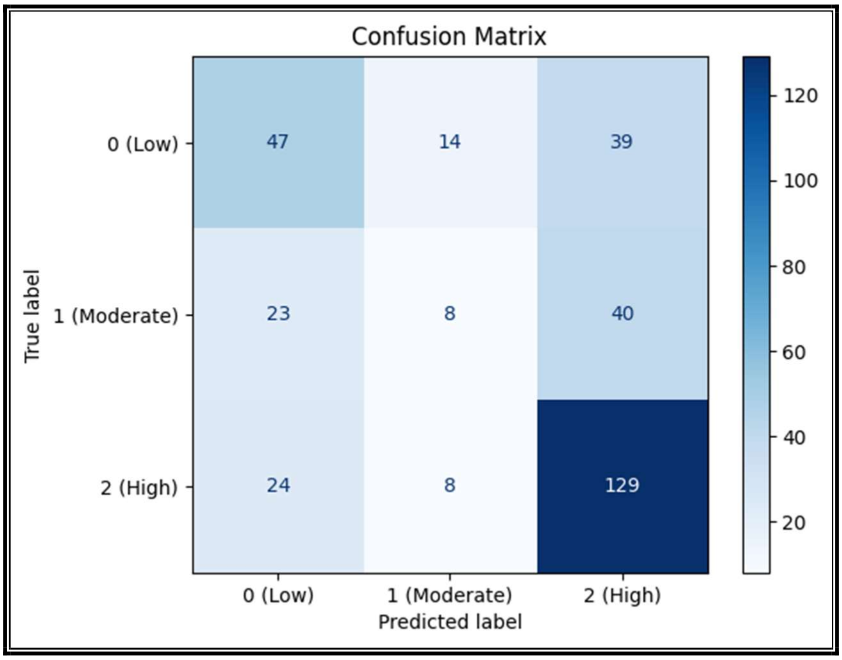


Figure 26: Confusion Matrix for the kNN classifier with k = 29

Report :

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.50 | 0.47 | 0.48 | 100 |
| 1 | 0.27 | 0.11 | 0.16 | 71 |
| 2 | 0.62 | 0.80 | 0.70 | 161 |
| accuracy | | | 0.55 | 332 |
| macro avg | 0.46 | 0.46 | 0.45 | 332 |
| weighted avg | 0.51 | 0.55 | 0.52 | 332 |

Figure 27: Classification Report for the kNN classifier with k = 29

```
The following metrics are obtained when the value of k = 29:
Accuracy: 55.42168674698795
Balanced Accuracy: 46.13060974542909
Precision: 50.83864303367316
Recall: 55.42168674698795
F1-Micro: 55.42168674698795
F1-Macro: 44.7379638642768
```

Figure 28: Various Metrics (%) for the kNN classifier with k = 29

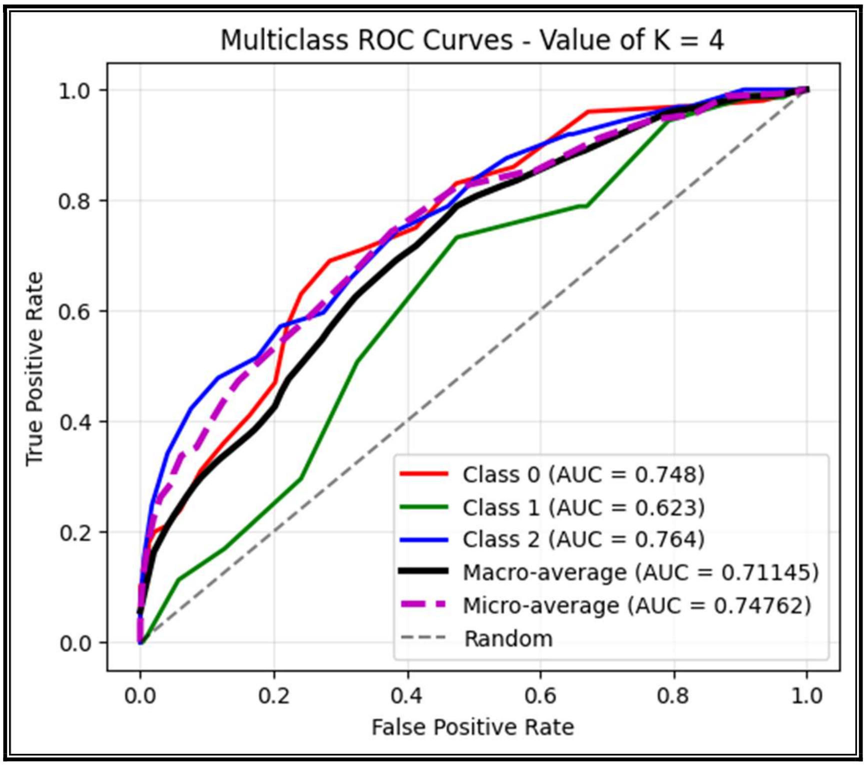


Figure 29: Multiclass ROC curve for the kNN classifier with k = 29

- The kNN classifier with the optimal k value reflects some of the earlier trends observed with the decision trees discussed above:

  1. High Recall value for Class 2, due to the relatively high number of True Positives that arise due to the high representation of Class 2 data in the given dataset.
  2. The kNN classifier reports an accuracy close to 55%, which is close to the one observed in the case of the decision trees above. Thus, in terms of accuracy, the decision trees and the kNN classifier perform similarly when applied to the Validation dataset.
  3. A relatively low value of AUC is observed for Class 1. However, the values of AUC for Classes 0 and 2 are comparable, which is a departure from what was observed in the case of the decision trees. The average values of AUC continue to be around 0.75.

h.

- The performance of the decision tree with depth = 2 on the Test data is given as follows (additionally, the performance of the decision tree with depth = 4 on the same data is provided in the .ipynb file containing the relevant code) :

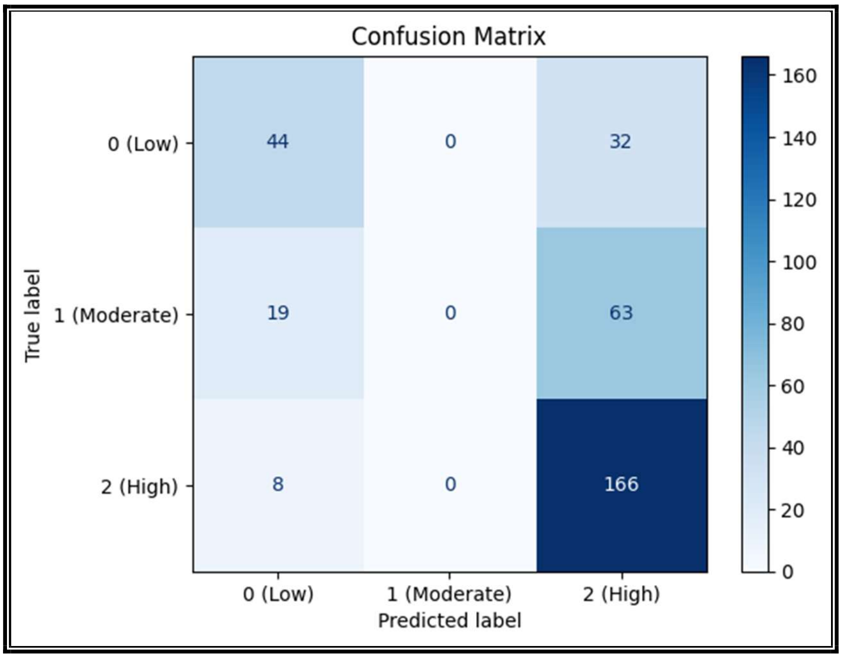


Figure 30: Confusion Matrix for the decision tree with depth = 2

```
Report :
              precision    recall  f1-score   support

           0       0.62      0.58      0.60        76
           1       0.00      0.00      0.00        82
           2       0.64      0.95      0.76       174

    accuracy                           0.63       332
   macro avg       0.42      0.51      0.45       332
weighted avg       0.48      0.63      0.54       332
```

Figure 31: Classification Report for the decision tree with depth = 2

```
The performance of the decision tree with depth = 2 on the test dataset is given as follows:
Accuracy: 63.25301204819277
Balanced Accuracy: 51.09901189756
Precision: 47.519656089145315
Recall: 63.25301204819277
F1-Micro: 63.25301204819277
F1-Macro: 45.39526155289702
```

Figure 32: Various metrics for the decision tree with depth = 2

- The performance of the kNN classifier with k = 29 on the Test data is given as follows:

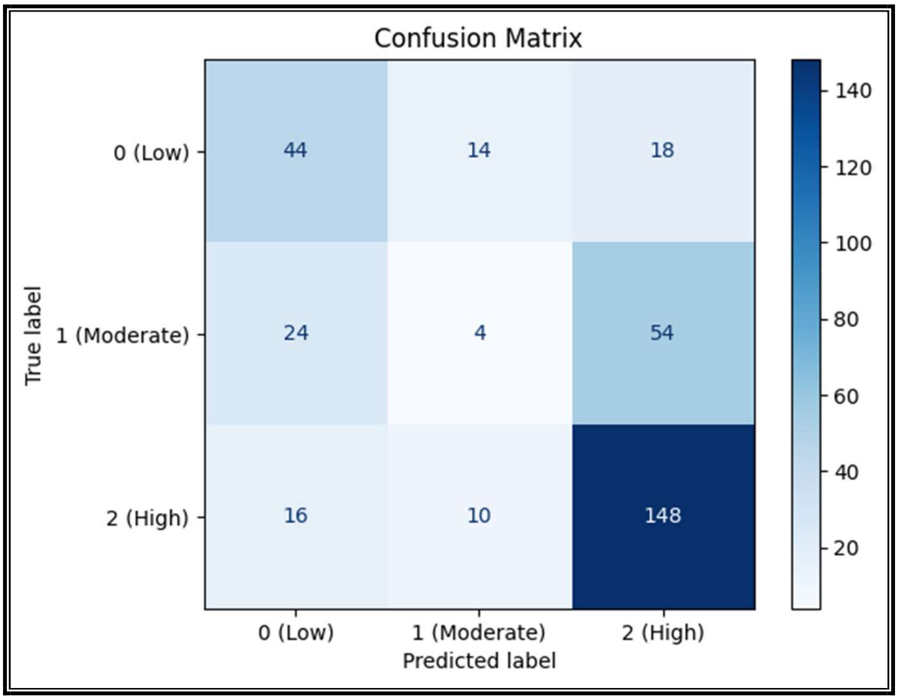


Figure 34: Confusion Matrix for the kNN classifier with k = 29

Report :

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.62 | 0.58 | 0.60 | 76 |
| 1 | 0.00 | 0.00 | 0.00 | 82 |
| 2 | 0.64 | 0.95 | 0.76 | 174 |
| accuracy | | | 0.63 | 332 |
| macro avg | 0.42 | 0.51 | 0.45 | 332 |
| weighted avg | 0.48 | 0.63 | 0.54 | 332 |

Figure 35: Classification Report for the kNN classifier with k = 29

```
The performance of the decision tree with depth = 2 on the test dataset is given as follows:
Accuracy: 59.036144578313255
Balanced Accuracy: 49.27675229565363
Precision: 50.776612945287646
Recall: 59.036144578313255
F1-Micro: 59.036144578313255
F1-Macro: 45.79987694200893
```

Figure 36: Various metrics for the kNN classifier with k = 29

- Thus, with an accuracy of 63%, the decision tree with depth = 2 performs better than the kNN classifier with k = 29, which has an accuracy of 59%.
- However, the decision tree with depth = 2 cannot classify elements into Class 1, which corresponds to the label ‘Moderate’. Moreover, the decision tree with depth = 4 (as shown in the .ipynb file containing the relevant code) has an accuracy of 57%, which is comparable to the one obtained by using the kNN classifier with k = 29.
- Thus, if we compare the decision tree with depth = 4 and the kNN classifier with k = 29, we conclude the kNN classifier to be the better one among the two in terms of accuracy.
- However, it must be noted that the ‘imbalance’ in the given dataset due to the preponderance of the values belonging to Class 2, which corresponds to the label ‘High’ could also be the reason behind the results obtained.

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